

Analysis & Intelligence, and AI

Not everything is AI

Dr. Patricia Ryser-Welch - Sept 2025



Background



Analysis

Analysis (pl.: analyses) is the process of breaking a complex topic or substance into smaller parts in order to gain a better understanding of it.

Data Analysis

Data analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making. It is about turning raw data into actionable insights

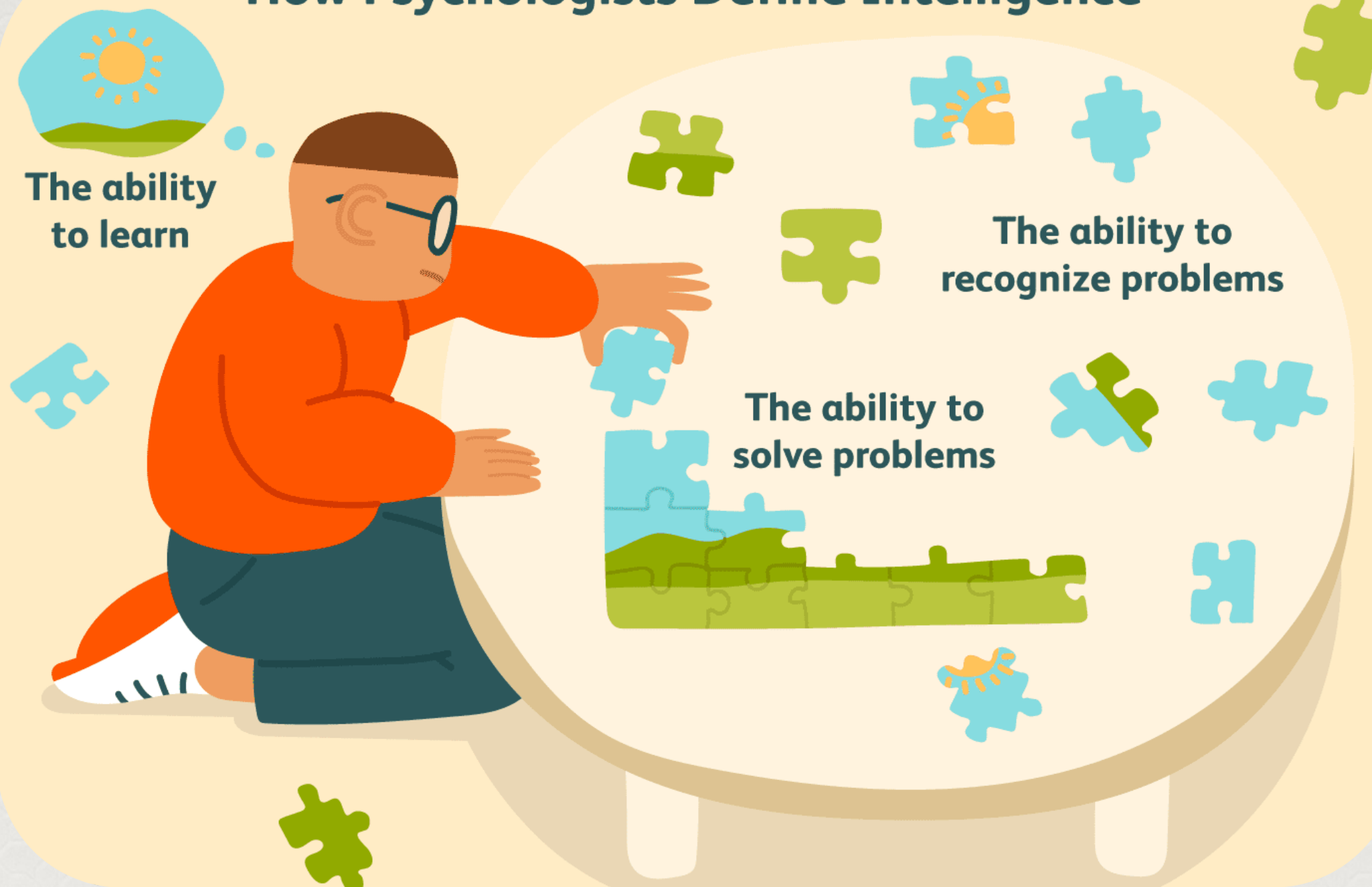
Intelligence

the ability to learn, understand, and make judgments or have opinions that are based on reason

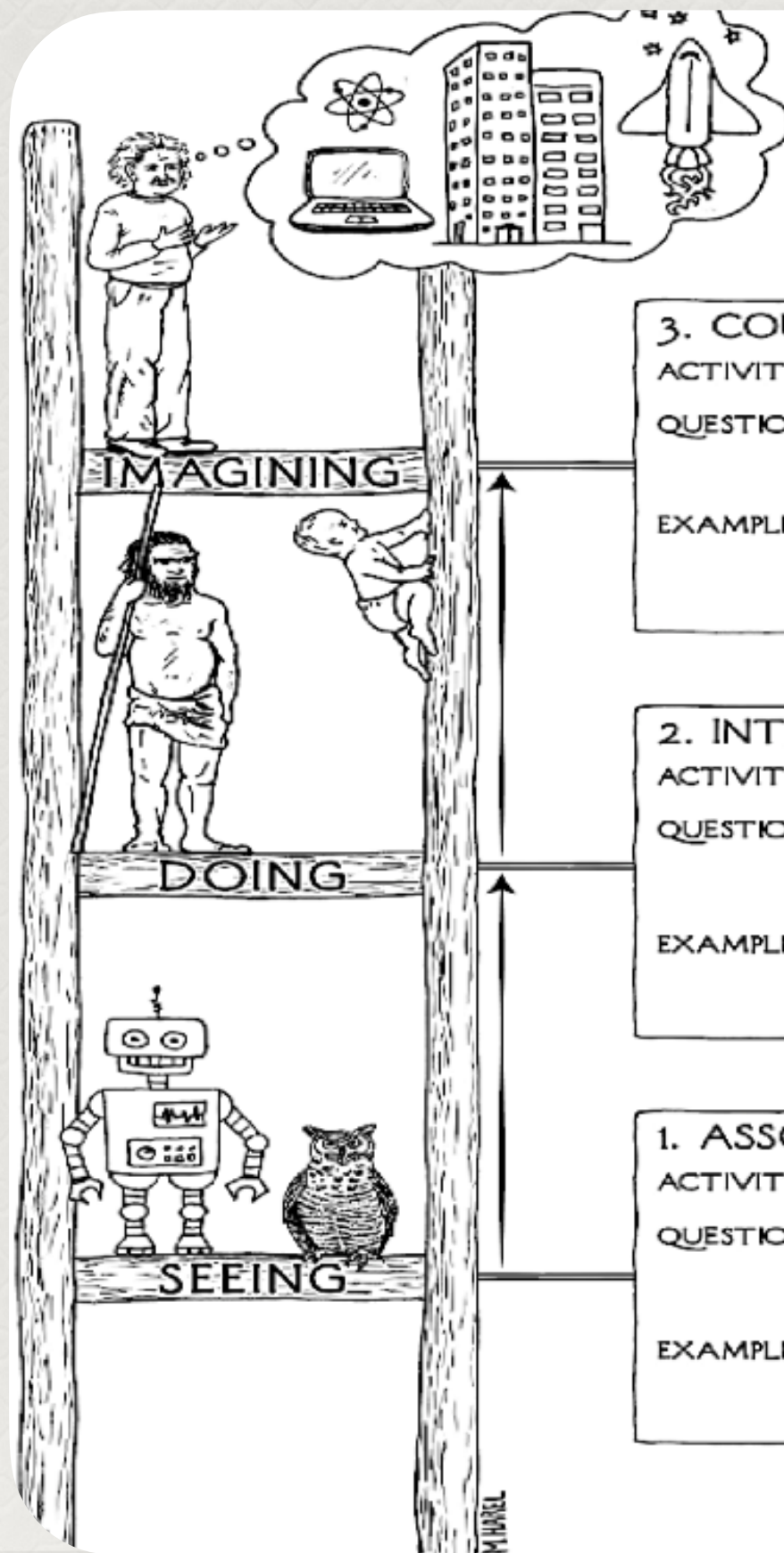
Business Intelligence

Business Intelligence (BI) refers to the strategies, methodologies, and technologies used by enterprises for the data analysis and management of business information to inform and improve decision-making.

How Psychologists Define Intelligence



<https://www.verywellmind.com/theories-of-intelligence-2795035>



3. COUNTERFACTUALS

ACTIVITY: Imagining, Retrospection, Understanding

QUESTIONS: *What if I had done ...? Why?*
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)

EXAMPLES: Was it the aspirin that stopped my headache?
Would Kennedy be alive if Oswald had not killed him? What if I had not smoked for the last 2 years?

2. INTERVENTION

ACTIVITY: Doing, Intervening

QUESTIONS: *What if I do ...? How?*
(What would Y be if I do X?
How can I make Y happen?)

EXAMPLES: If I take aspirin, will my headache be cured?
What if we ban cigarettes?

1. ASSOCIATION

ACTIVITY: Seeing, Observing

QUESTIONS: *What if I see ...?*
(How are the variables related?
How would seeing X change my belief in Y?)

EXAMPLES: What does a symptom tell me about a disease?
What does a survey tell us about the election results?

Artificial Intelligence

"to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."

John McCarthy (1956) 

Artificial Intelligence

"It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable."

John McCarthy (2004)

Artificial Intelligence

"products and services that are 'adaptable' and 'autonomous'"

United Kingdom Government (2023)

Learning
to gain knowledge or skill by studying, from experience, from being taught

Machine Learning

"field of study that gives computers the ability to learn without being explicitly programmed."

Arthur L. Samuel (1959) 

Machine Learning

a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalise to unseen data, and thus perform tasks without explicit instructions

John Koza (1996)

Machine learning

A type of AI by which computers find patterns in data or solve problems automatically without having to be explicitly programmed. All AI in current use is built using ML techniques.

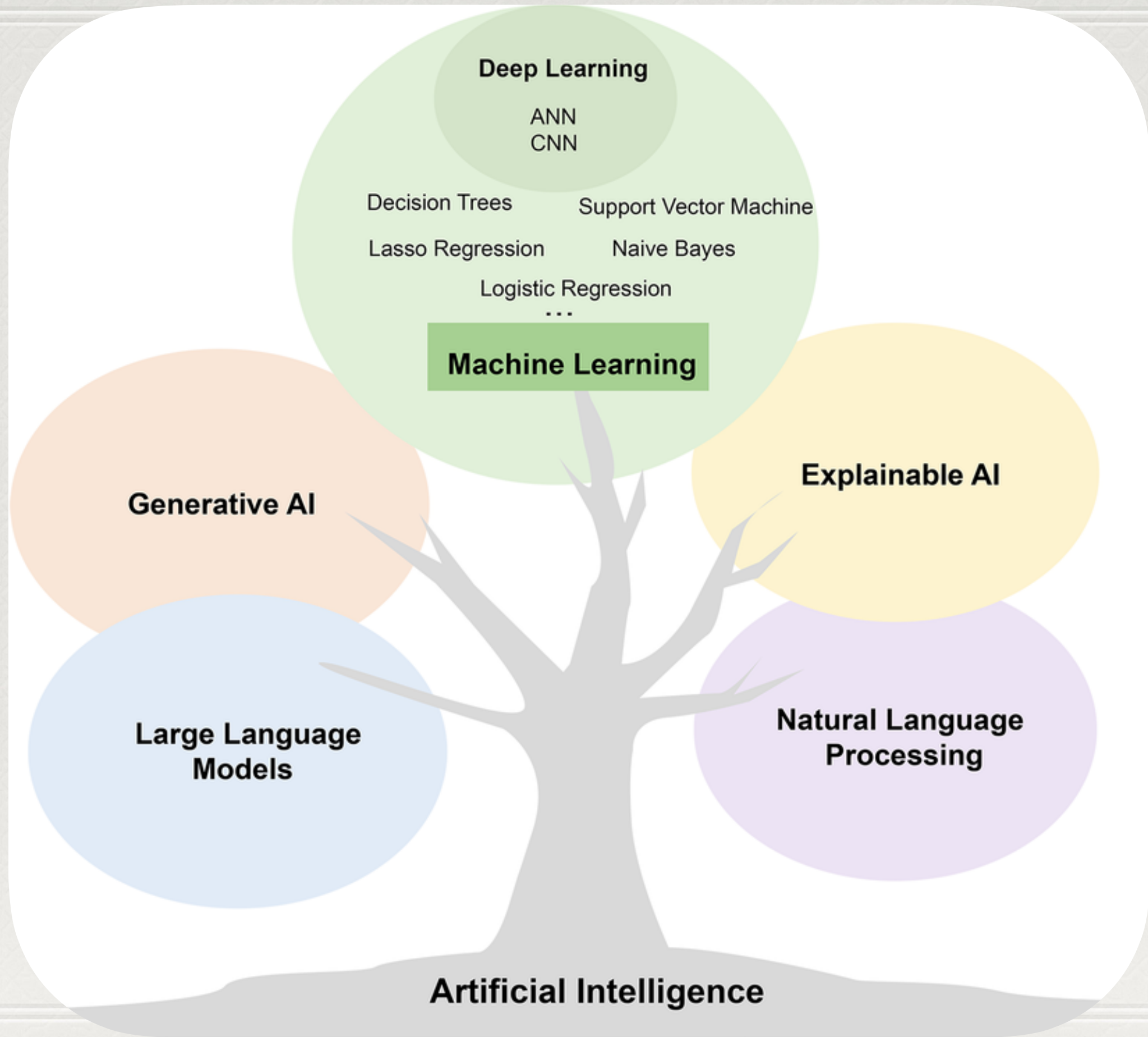
United Kingdom Government (2023) 

Reasoning

Reason is the capacity of consciously applying logic by drawing valid conclusions from new or existing information, with the aim of seeking truth.

Reasoning in AI

Reasoning in artificial intelligence (AI) refers to the mechanism of using available information to generate predictions, make inferences and draw conclusions. It involves representing data in a form that a machine can process and understand, then applying logic to arrive at a decision.

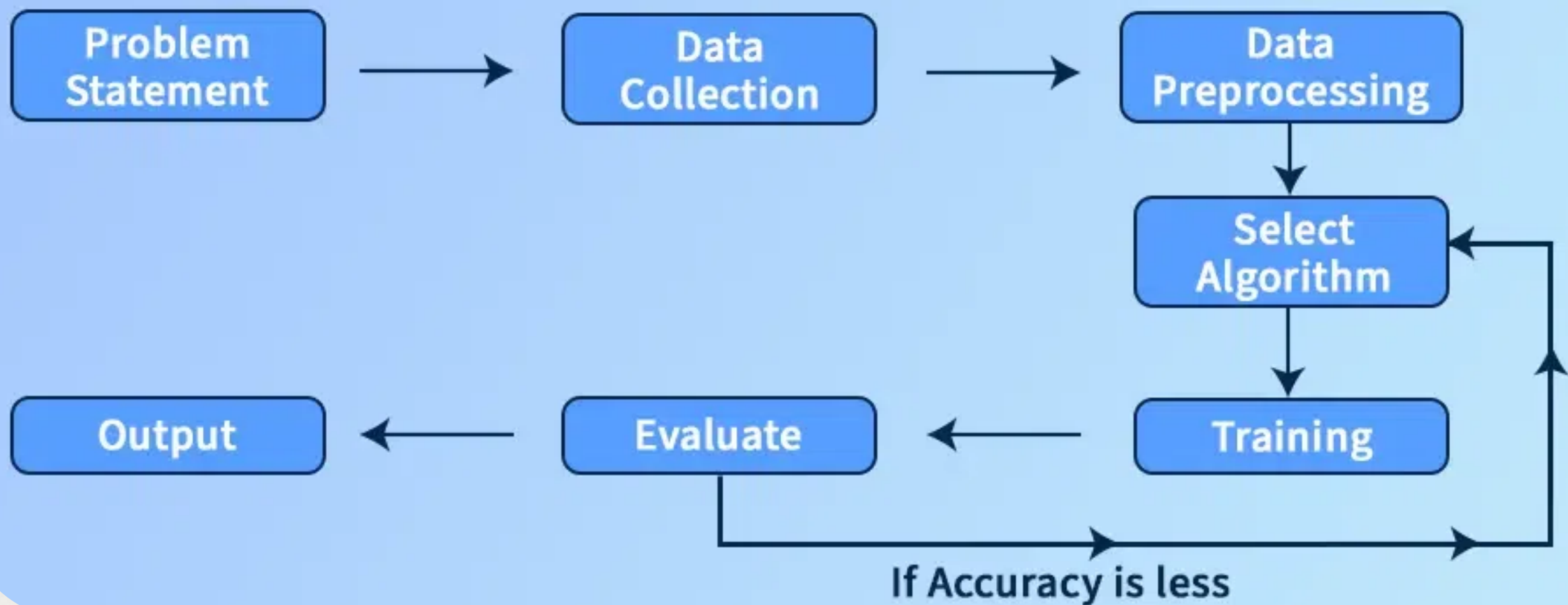


ML Techniques



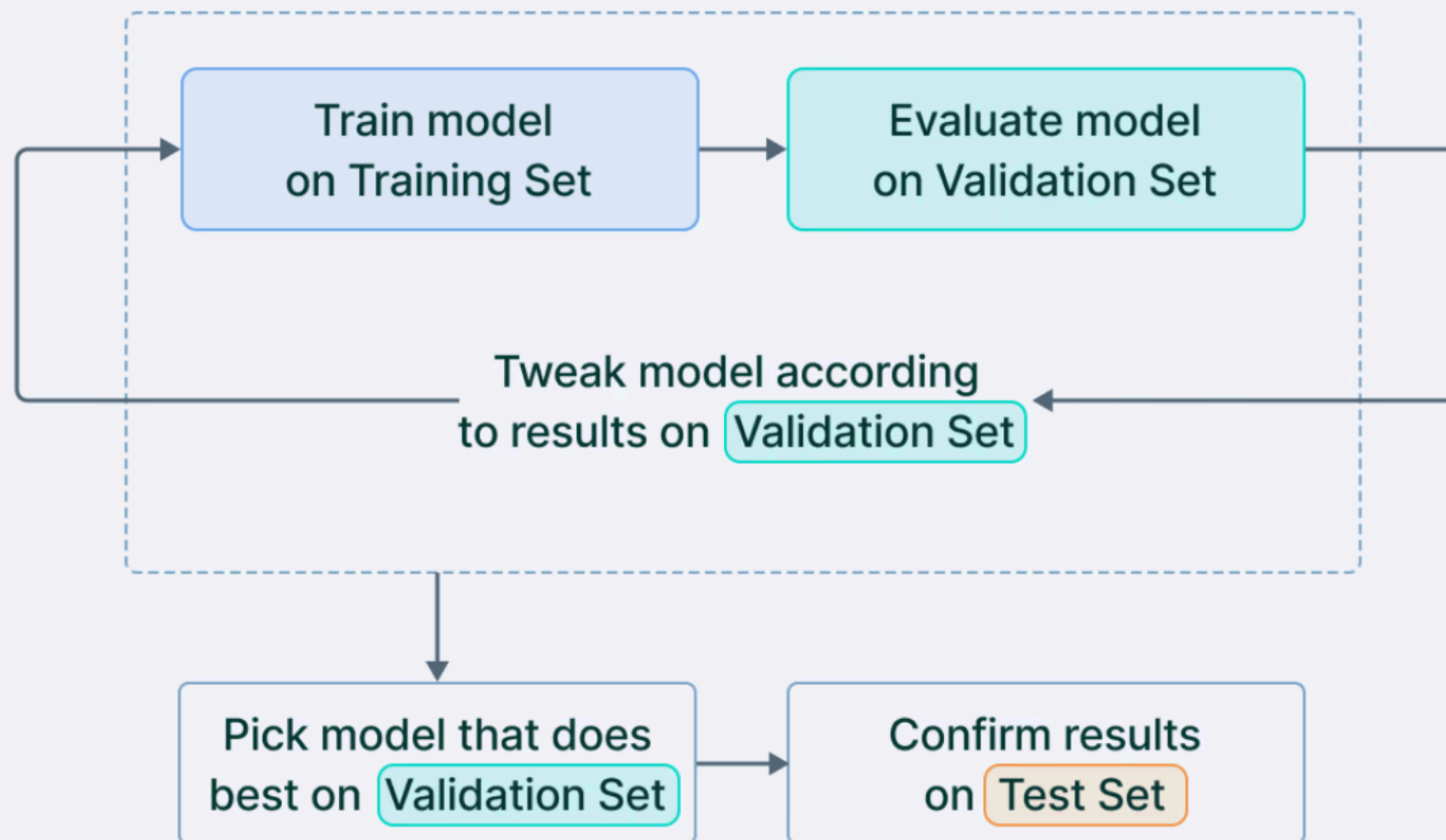
ML process

Machine Learning Model



AI and ML are both data hungry

Training data/validation/test



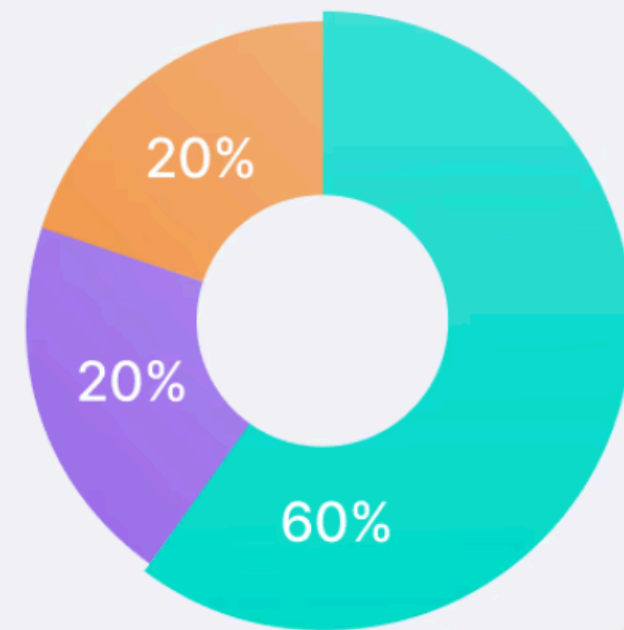
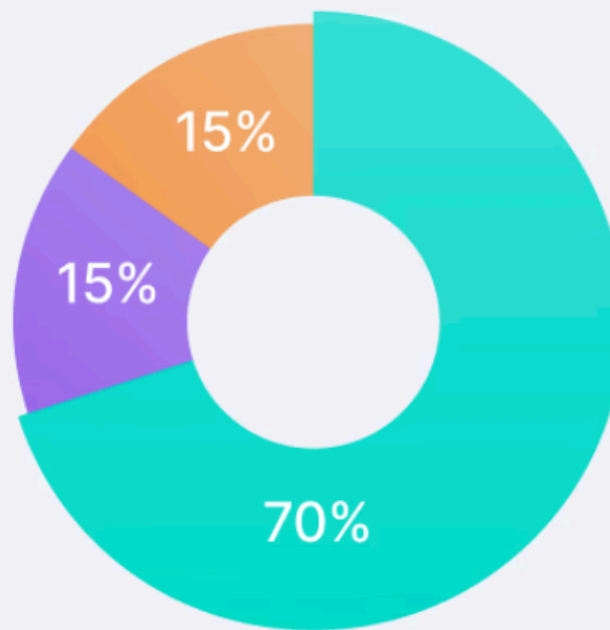
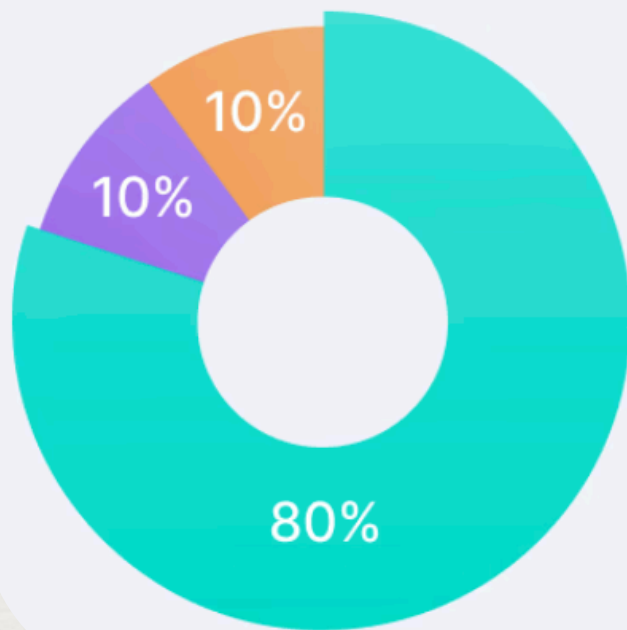
Most of the data is used for training

Data Training Needs

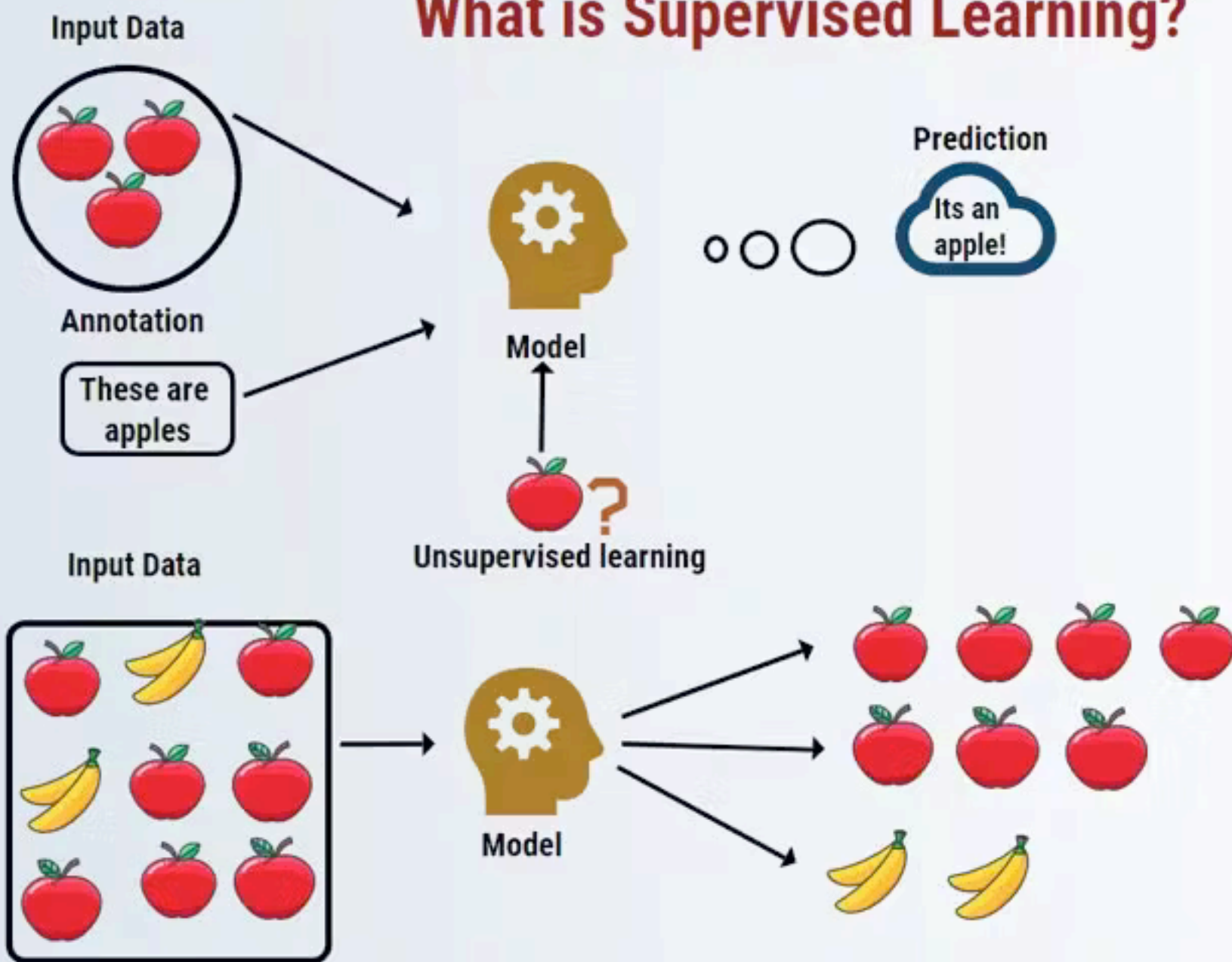
Training data

Validation data

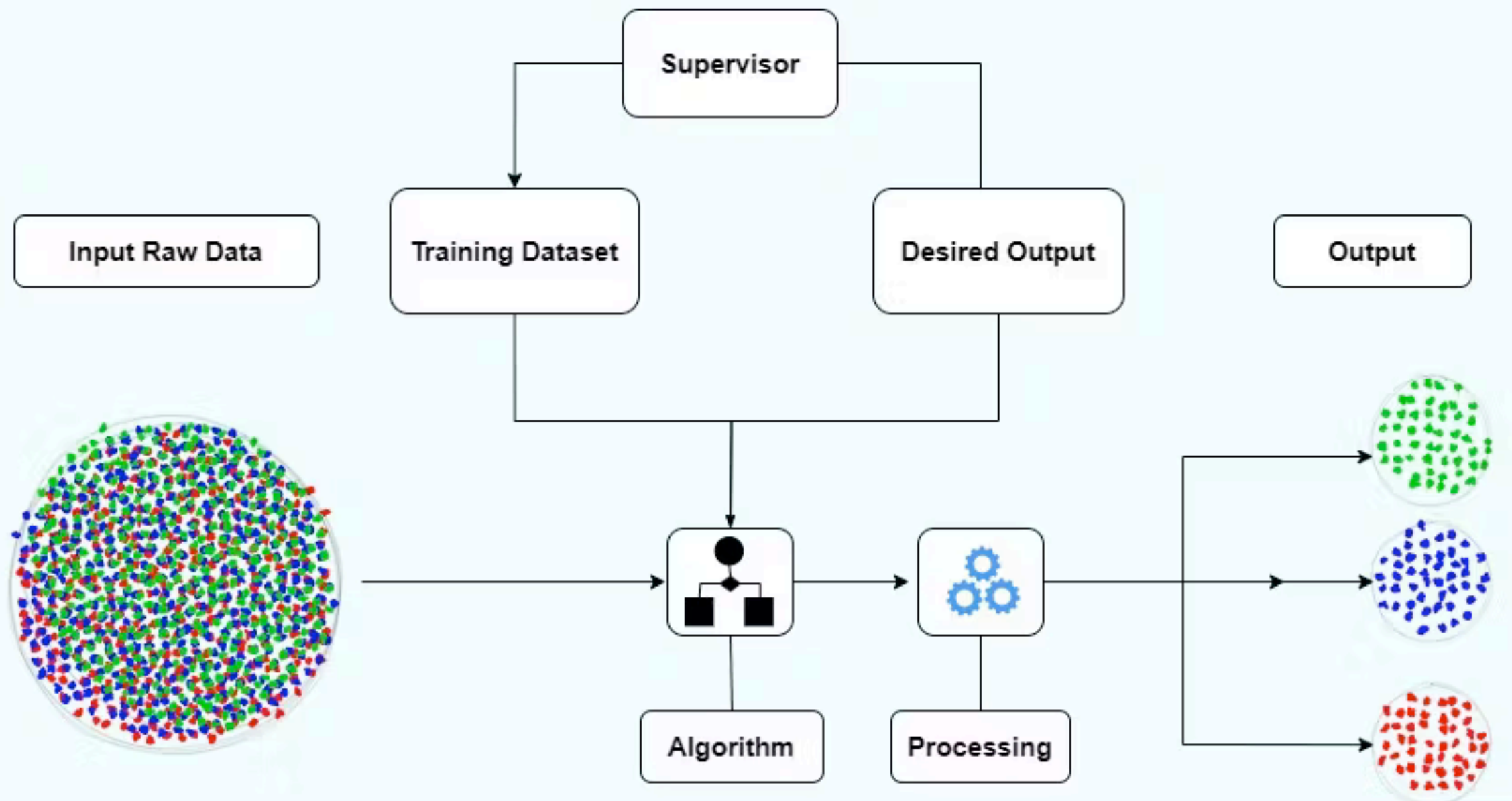
Test data



What is Supervised Learning?



Supervised Learning



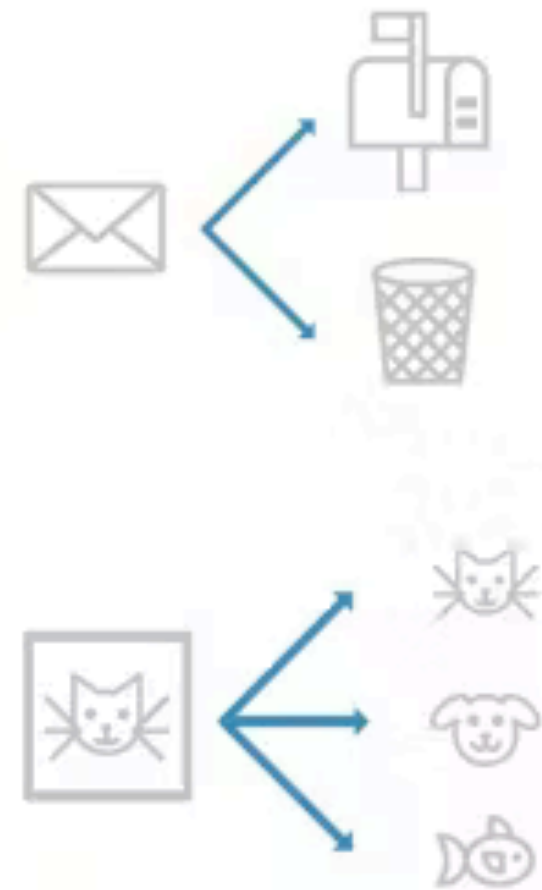
Form

Classification

Regression

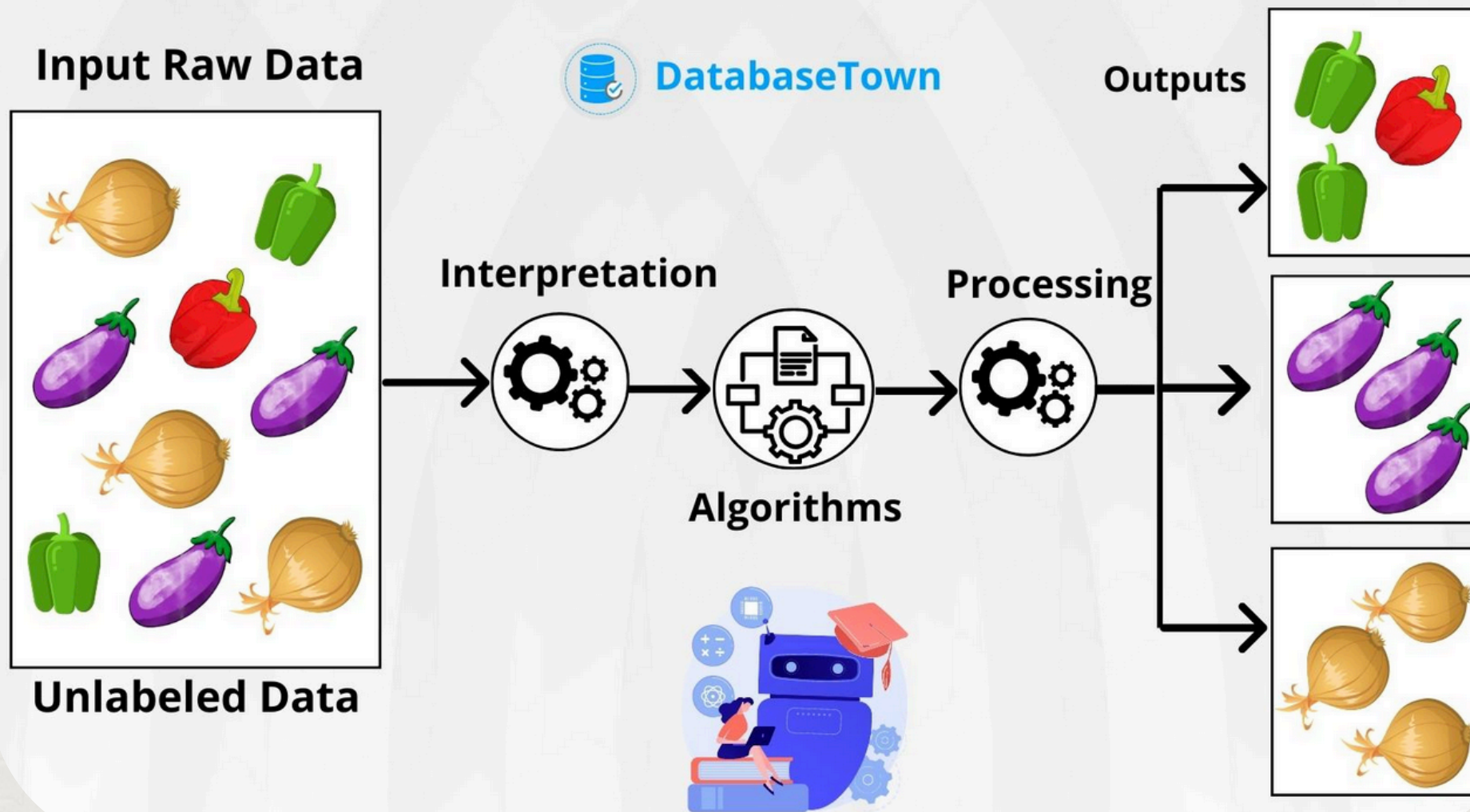
Binary

Multi Class

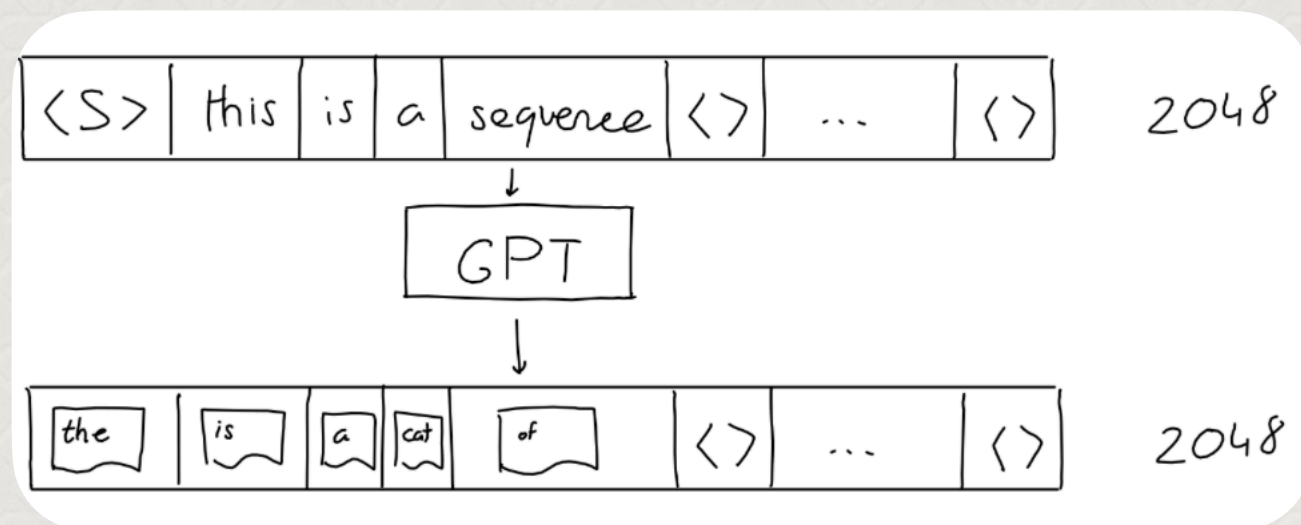
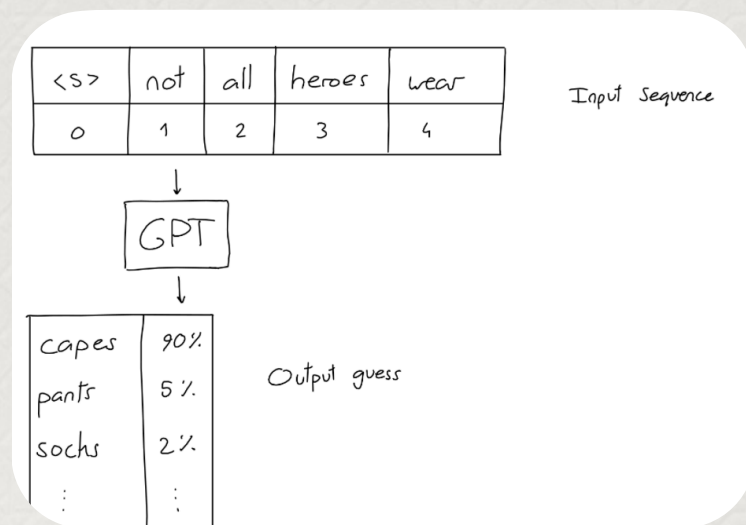


UNSUPERVISED LEARNING

Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data without any predefined outputs or target variables.

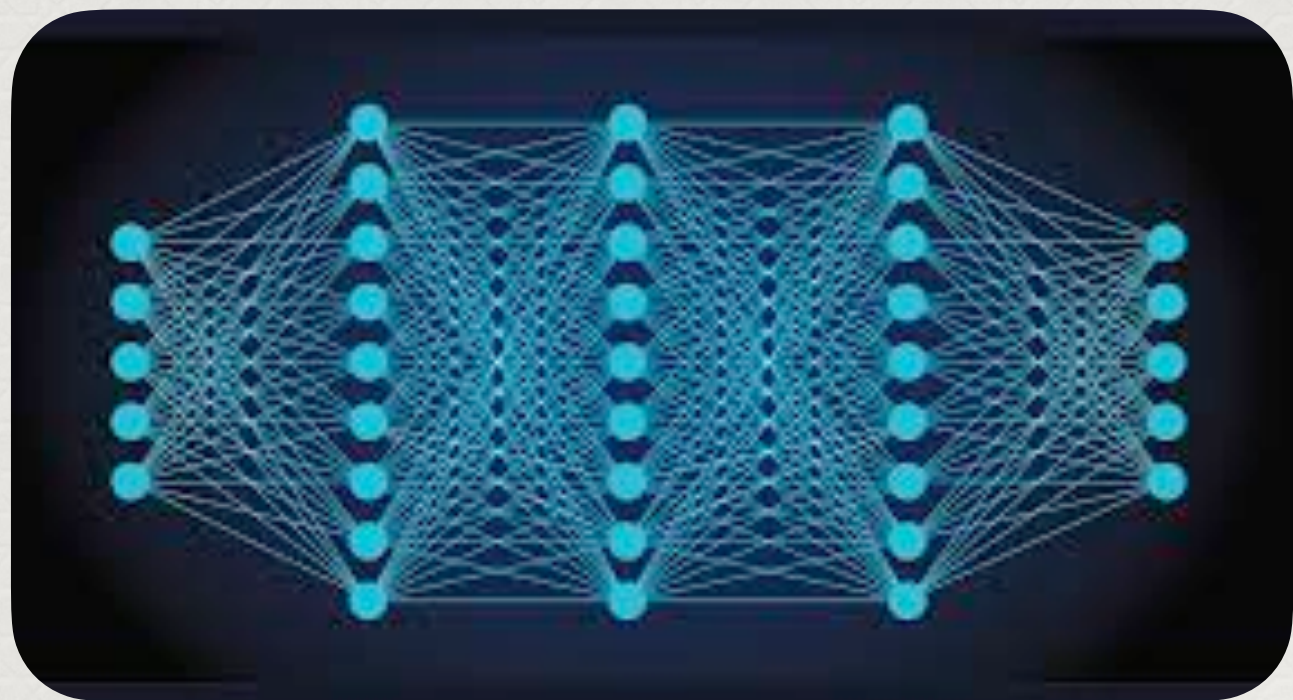


Simulating reasoning - with Generative Pre-trained Transformer

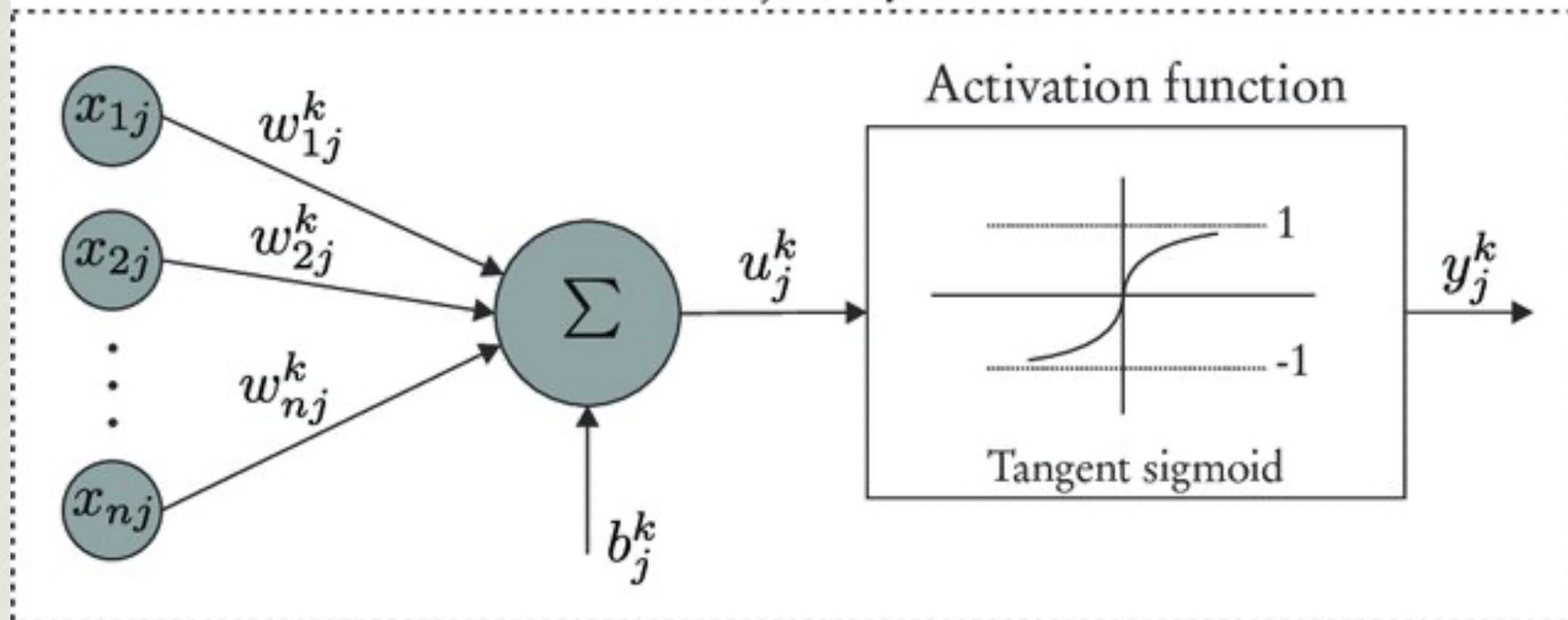


In and out - all sequences where the words are guessed

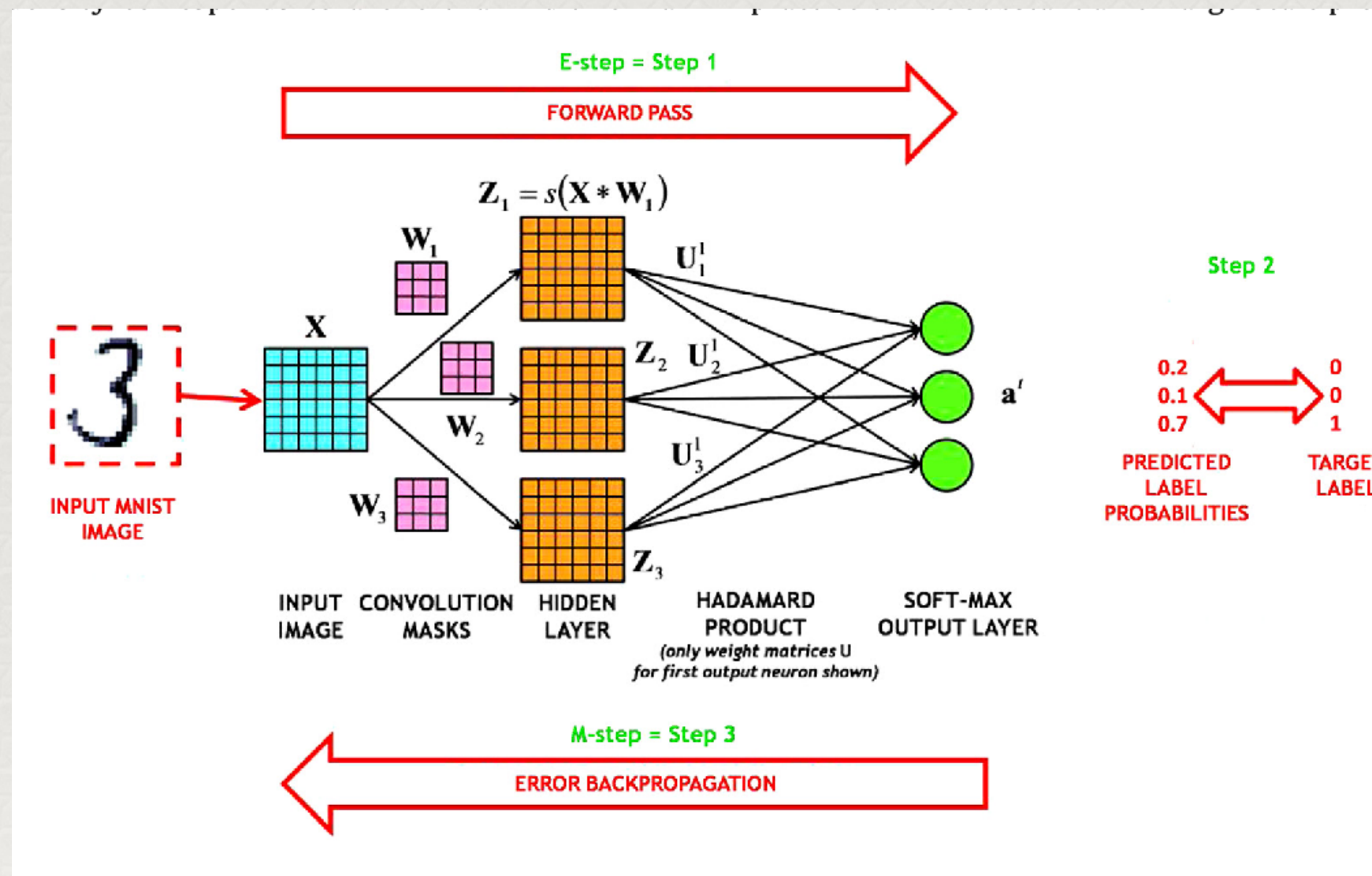
Neural AI



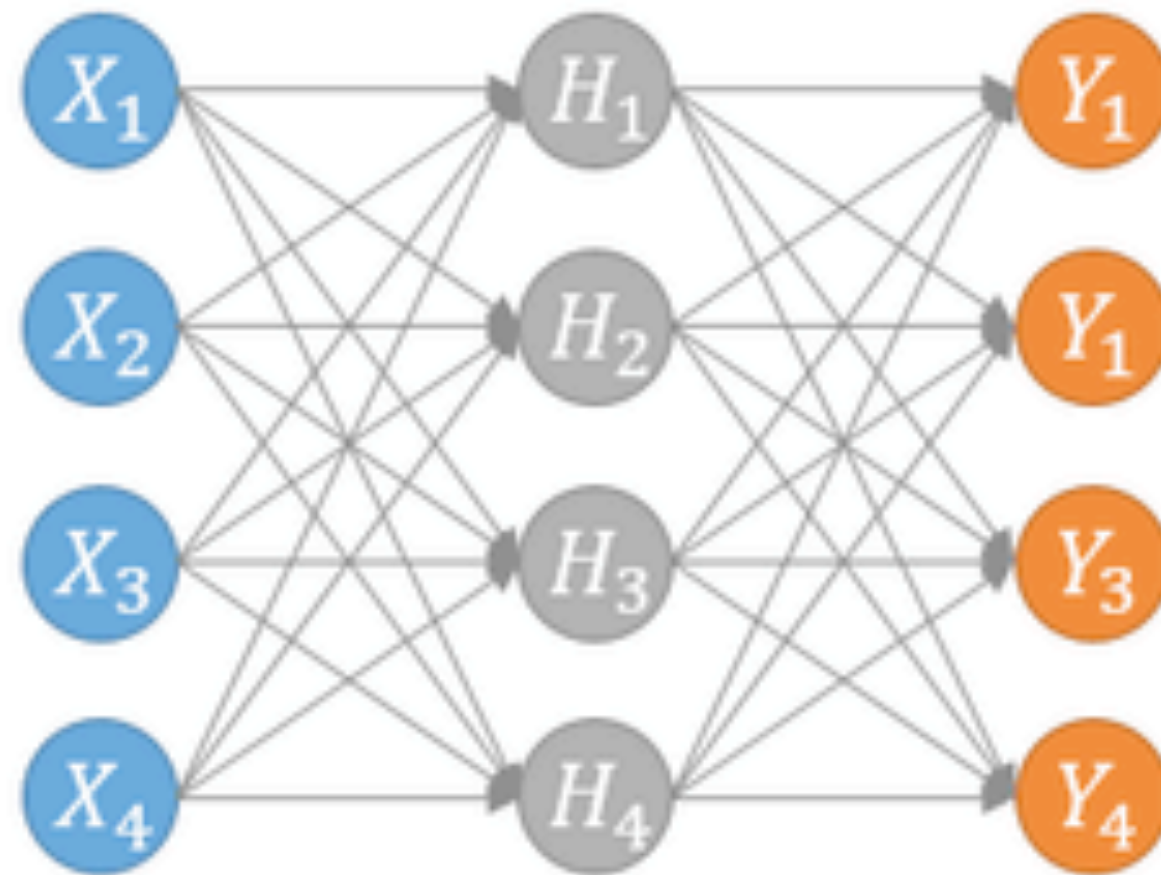
Neuron j in layer k



Forward and back propagation algorithm



(a) Typical ANN

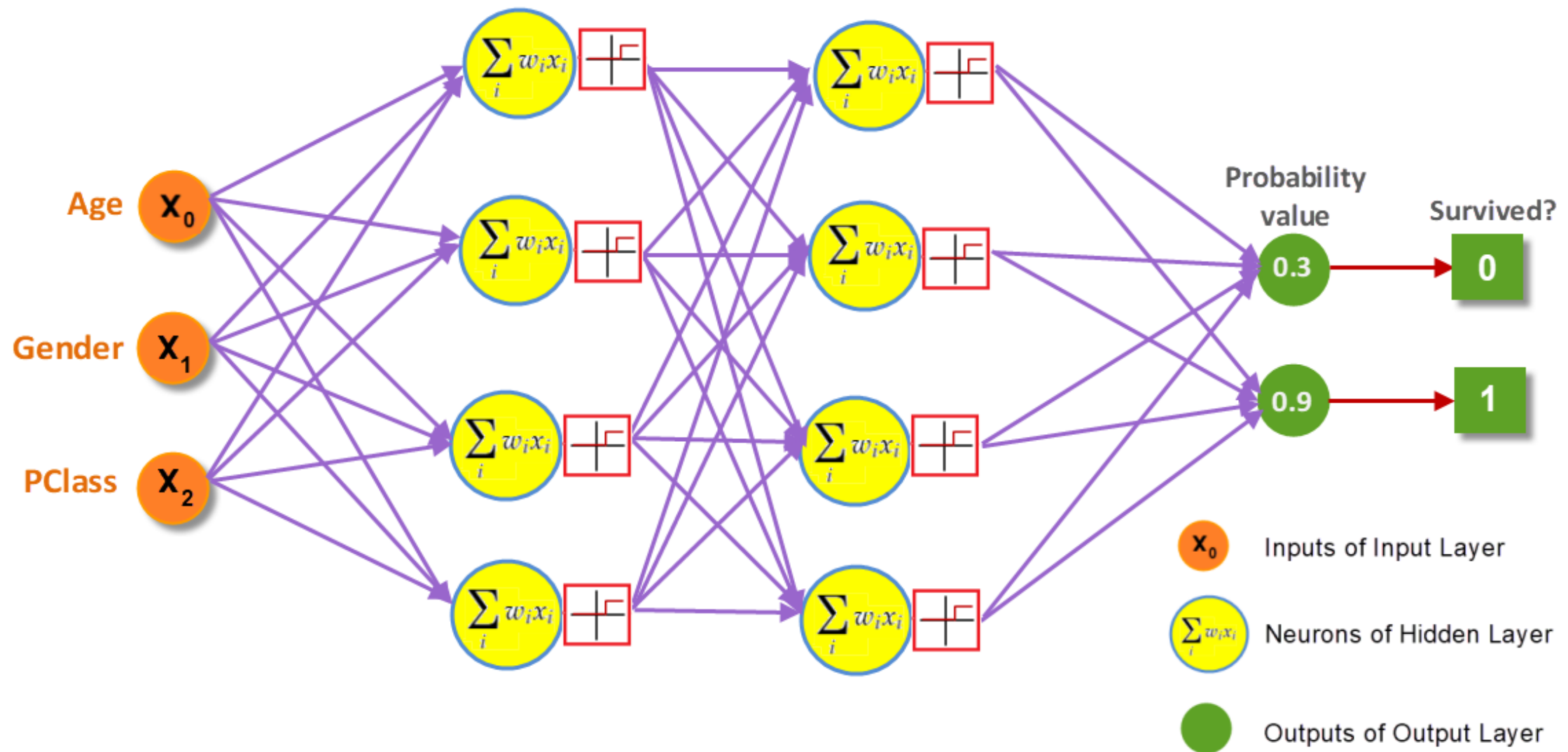


input
layer

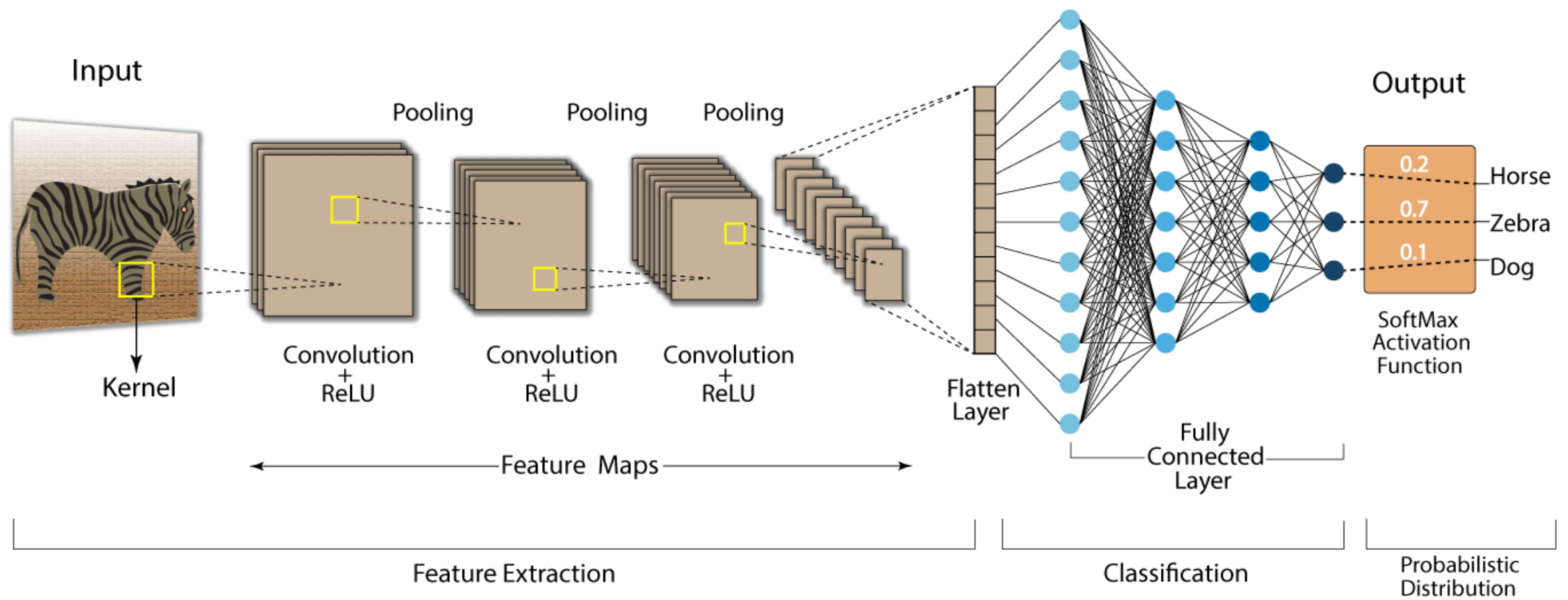
hidden
layer

output
layer

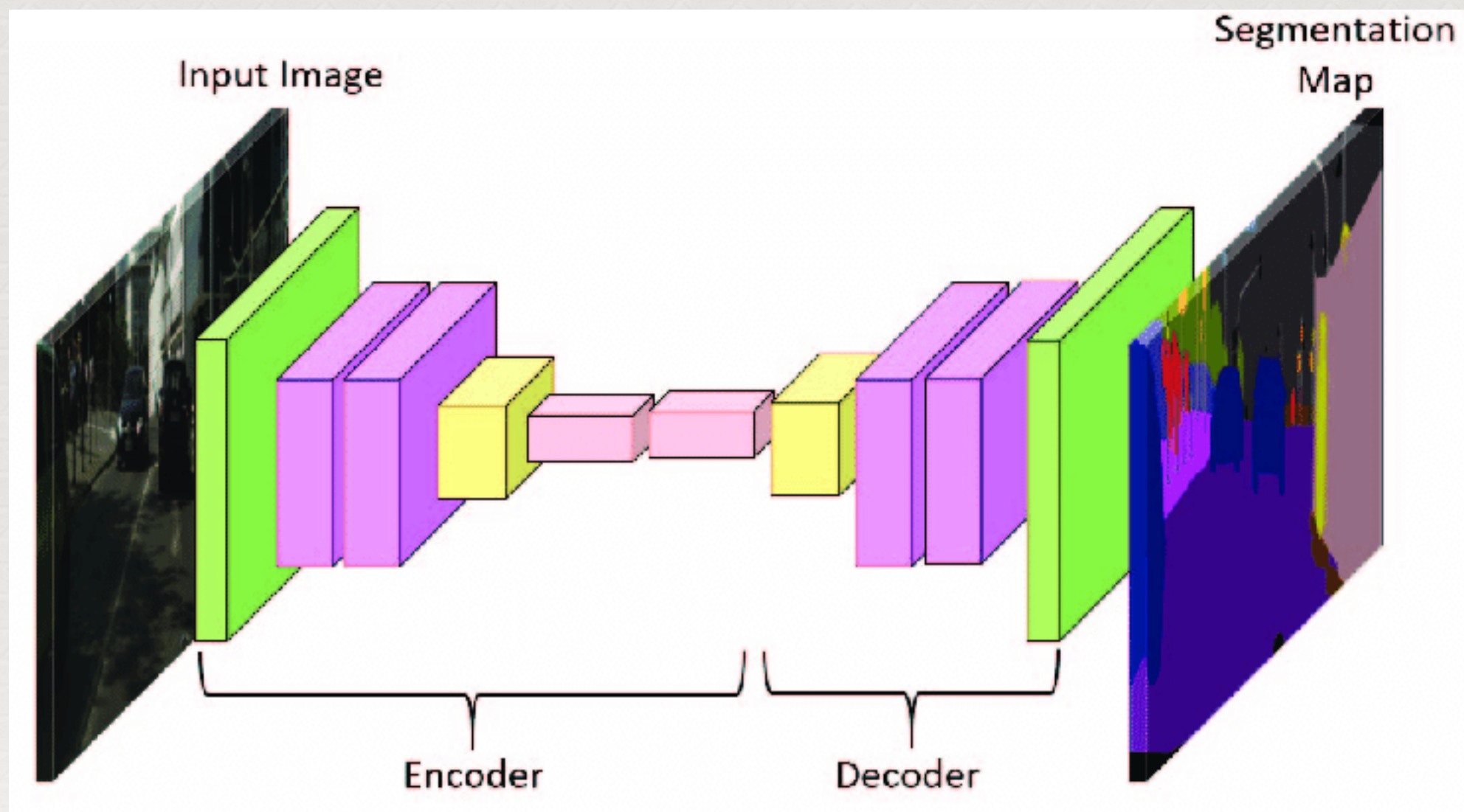
Predicting whether a passenger may have survived the Titanic Disaster?



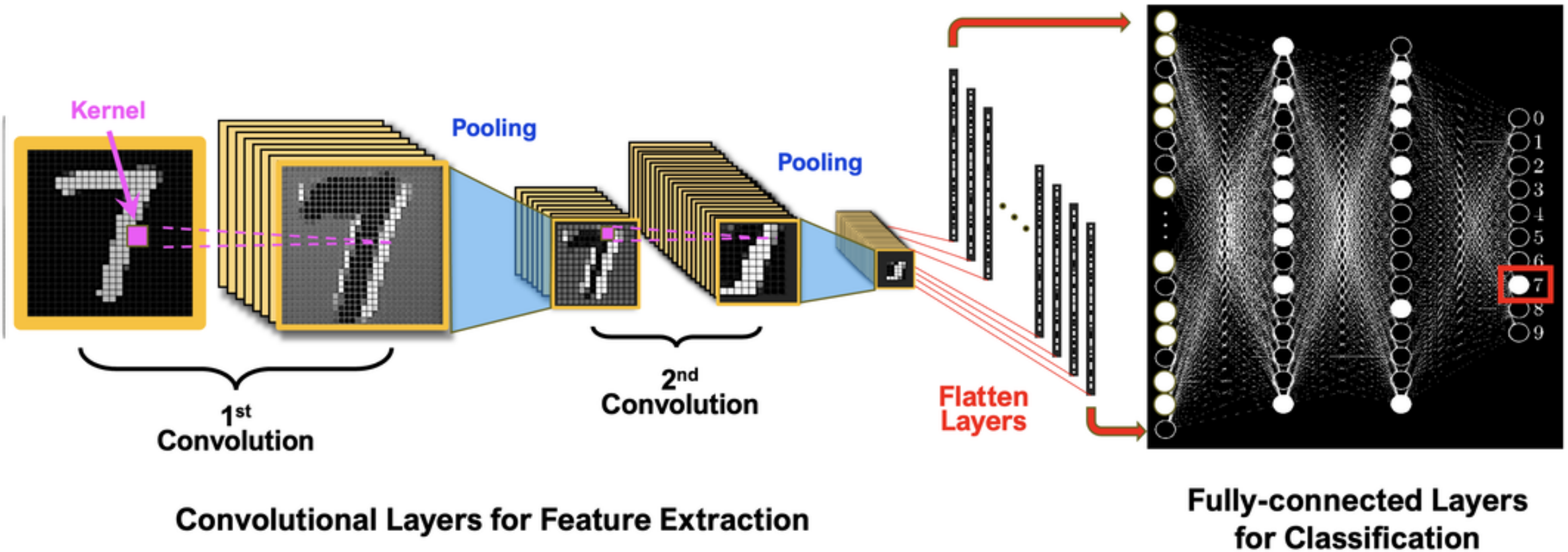
Convolution Neural Network (CNN)



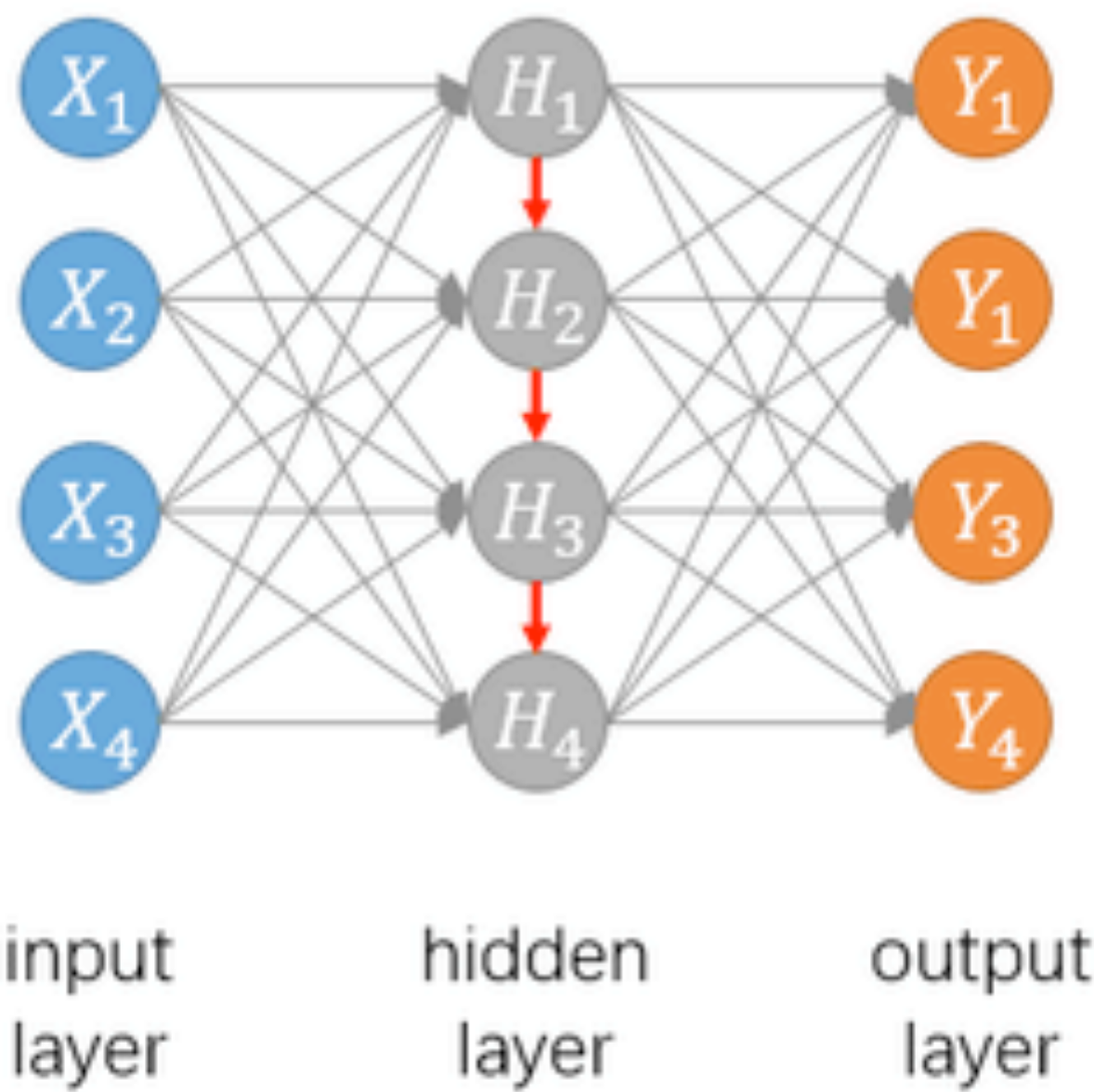
Images transformation

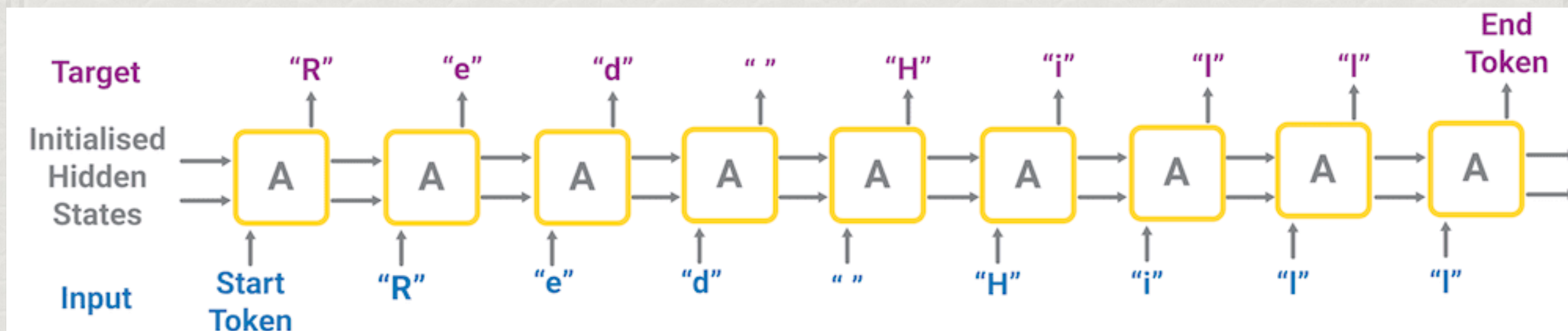


CNN - images classification

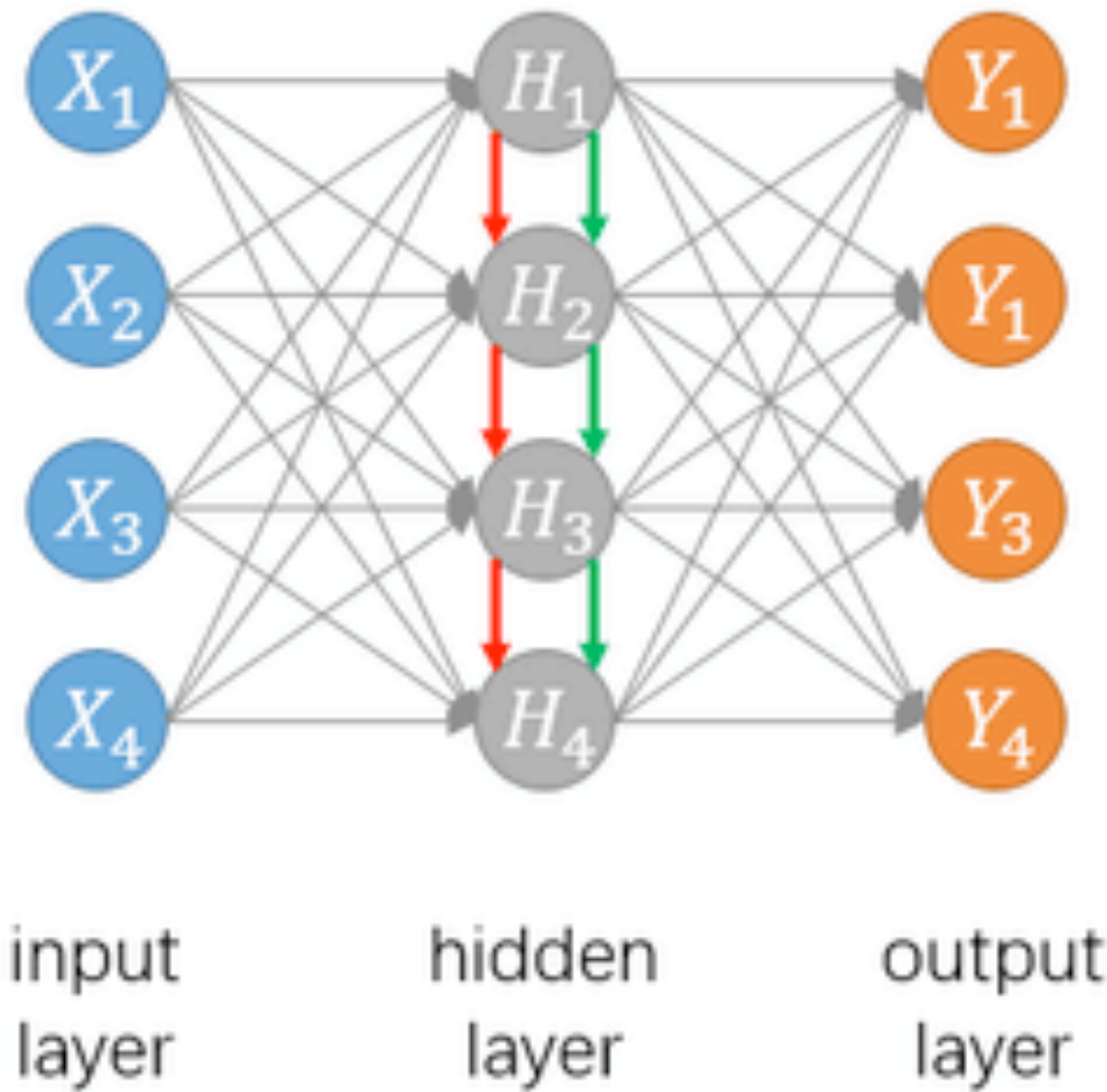


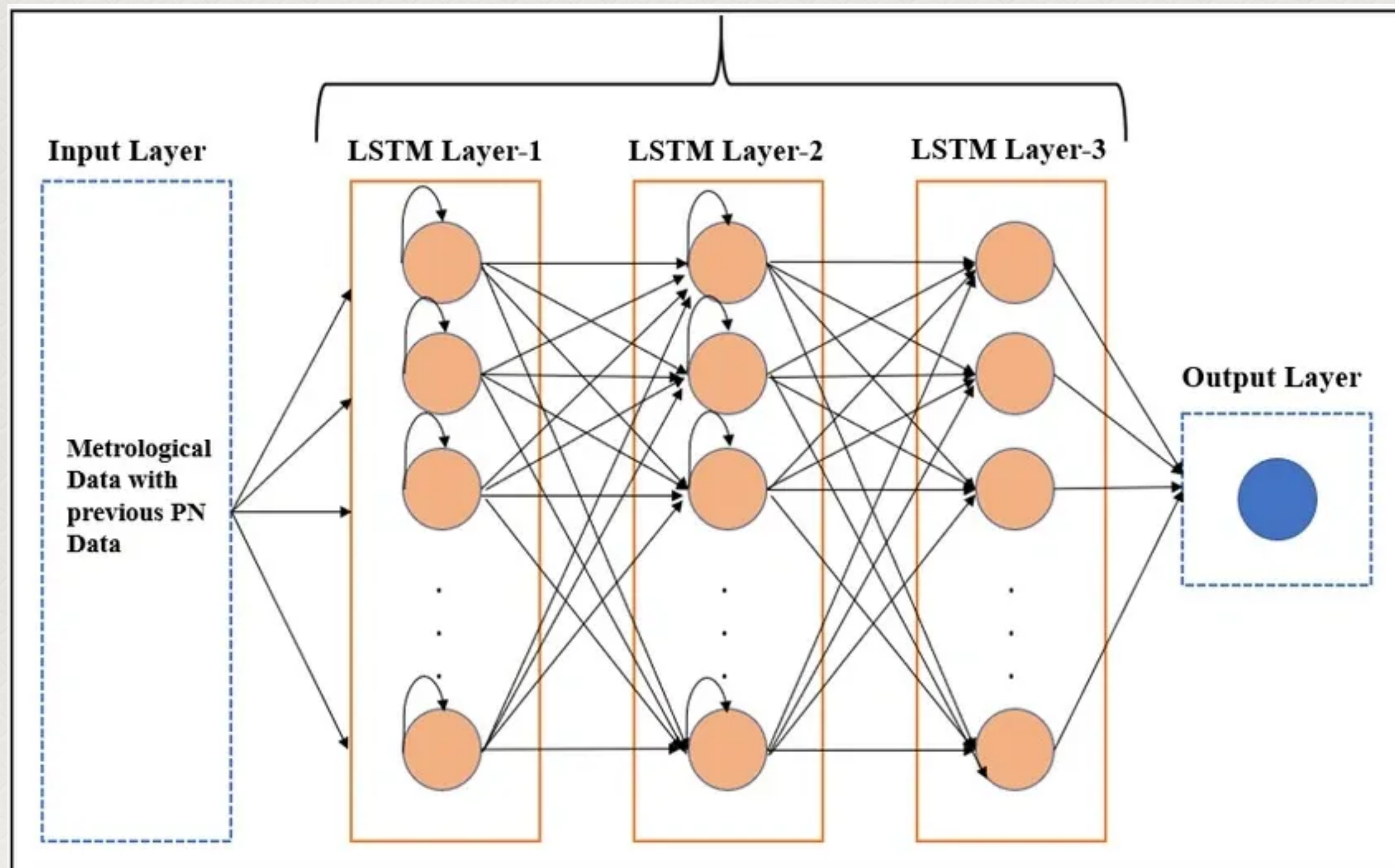
(b) RNN





(c) LSTM





An example of an LLMs - Chat GPT 3 - now obsolete

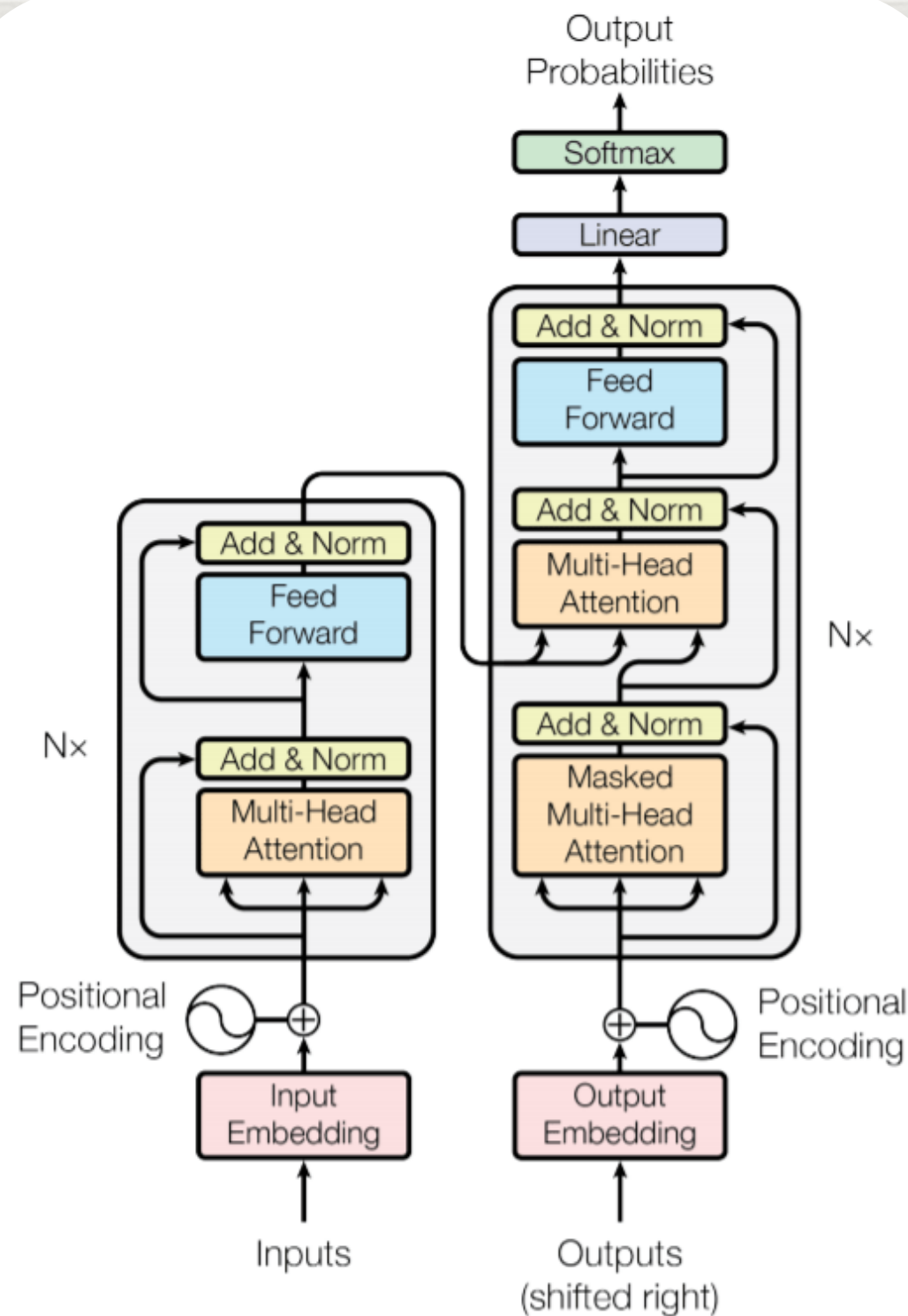
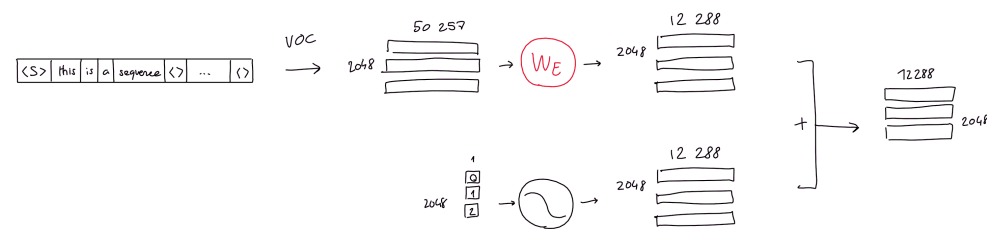


Figure 1: The Transformer - model architecture.

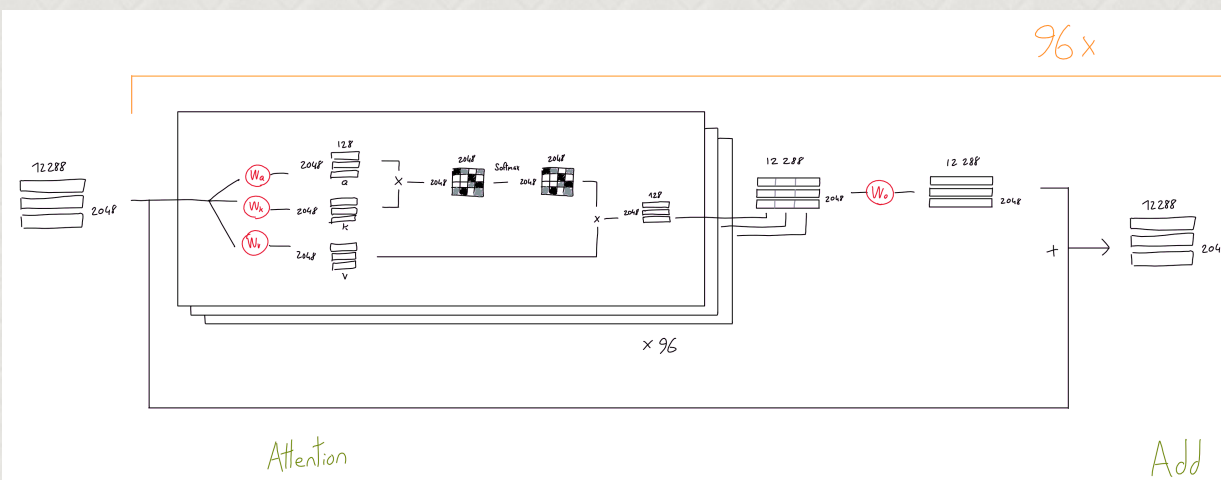
https://dugas.ch/artificial_curiosity/GPT_architecture.html

Full architecture - it is very complex!



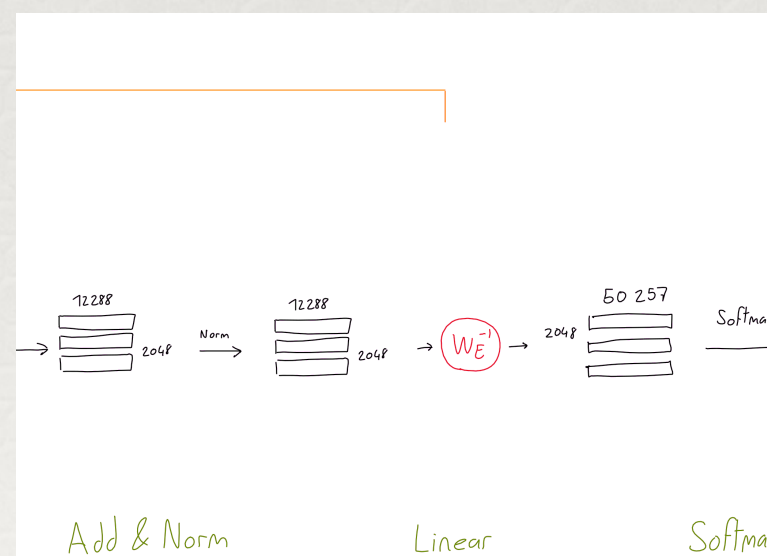
Encoding

Embedding



Attention

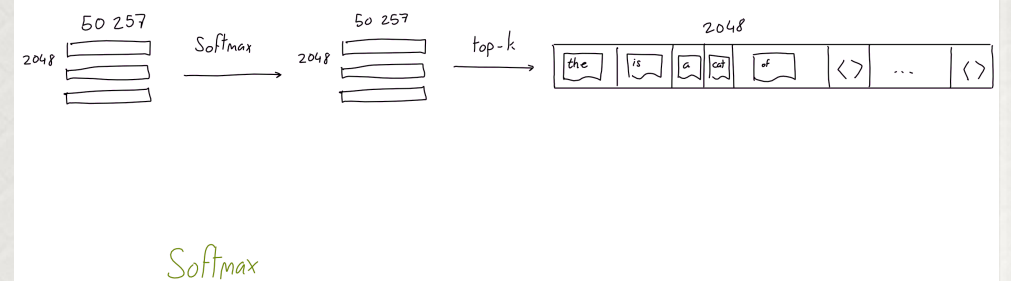
Add



Add & Norm

Linear

Softmax



Softmax

Encoding

The $\rightarrow$ $[0\ 0\ 0\ 0\ 1\ 0\ 0\ \dots]$

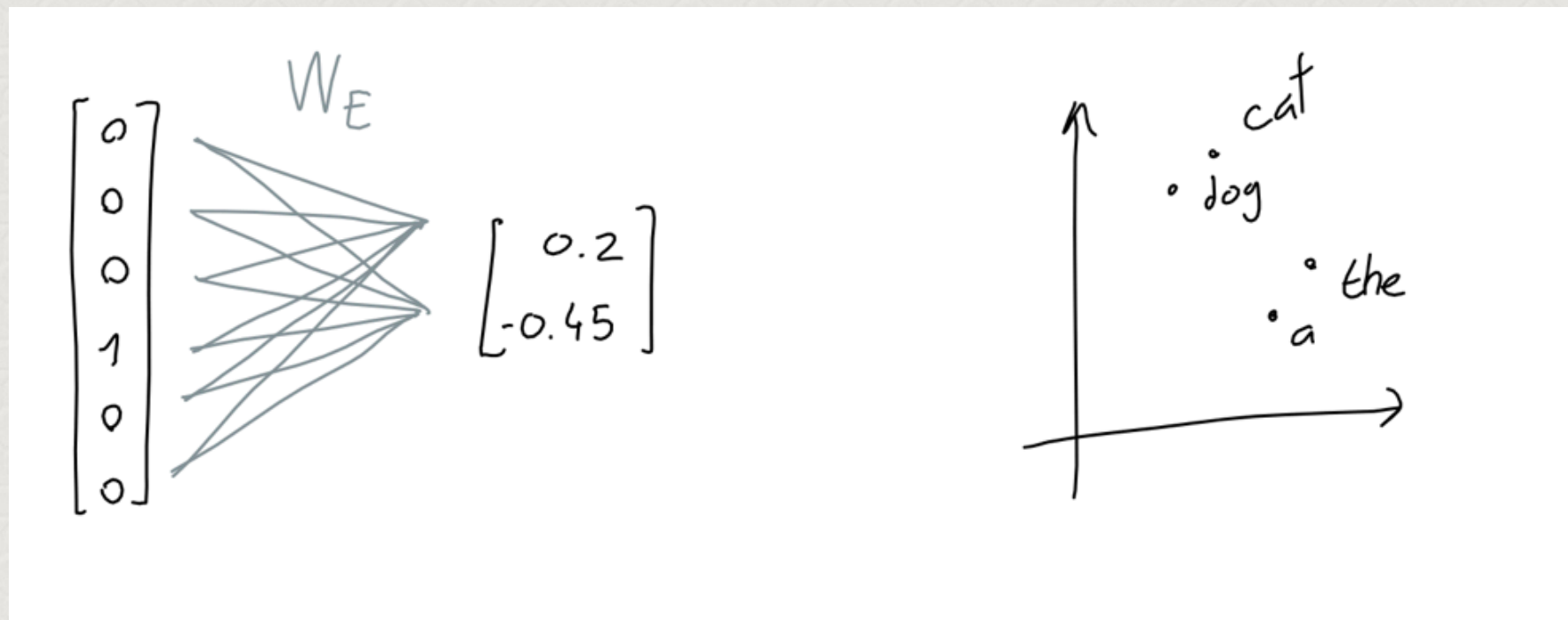
$\langle S \rangle$ | this | is | a | sequence | $\langle \rangle$ | ... | $\langle \rangle$

$\xrightarrow{\text{VOC}}$

50' 257

0	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 & \dots & 0 \\ \vdots & & & \dots & & & \\ 0 & 0 & 1 & 0 & 0 & \dots & 0 \end{bmatrix}$	$\langle S \rangle$
1		this
2		is
$\vdots$		
2048		$\langle \rangle$

Embeddings simplify words in a 2-dimensional space

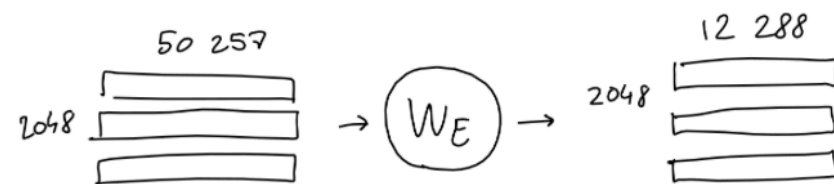


GPT can use 12,288 dimensions

In arithmetic terms, we multiply the 2048 x 50257 sequence-encodings matrix with the 50257 x 12288 embedding-weights matrix (learned) and end up with a 2048 x 12288 sequence-embeddings matrix.

$$\begin{array}{c} 50'257 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ \vdots \\ 2048 \end{matrix} \end{array} \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 1 & 0 & 0 & \dots & 0 \end{bmatrix} \times \begin{array}{c} 12\ 288 \\ 50'257 \\ W_E \end{array} = \begin{array}{c} 12\ 288 \\ 2048 \end{array} \begin{bmatrix} 0.1 & \dots & -0.2 \\ \vdots & \ddots & \vdots \\ 0.3 & \dots & -2.5 \end{bmatrix}$$

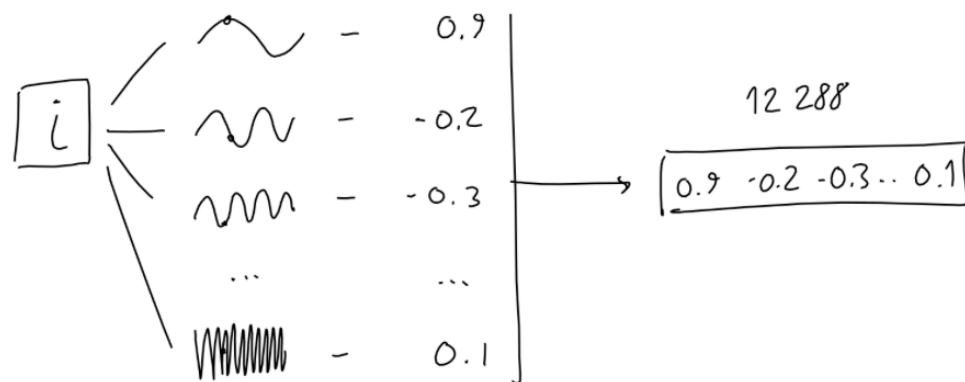
(A large curly brace is drawn under the entire equation.)



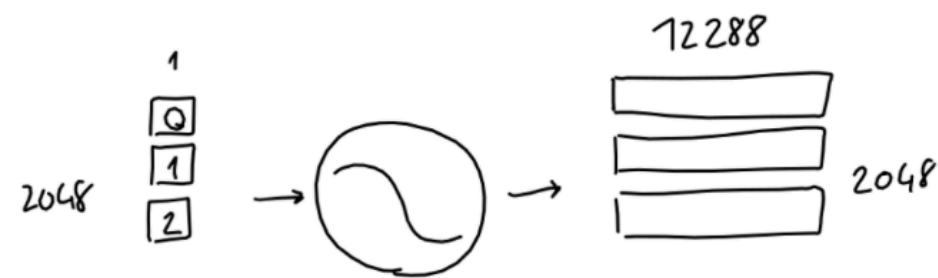
Positional encoding

Positional Encoding

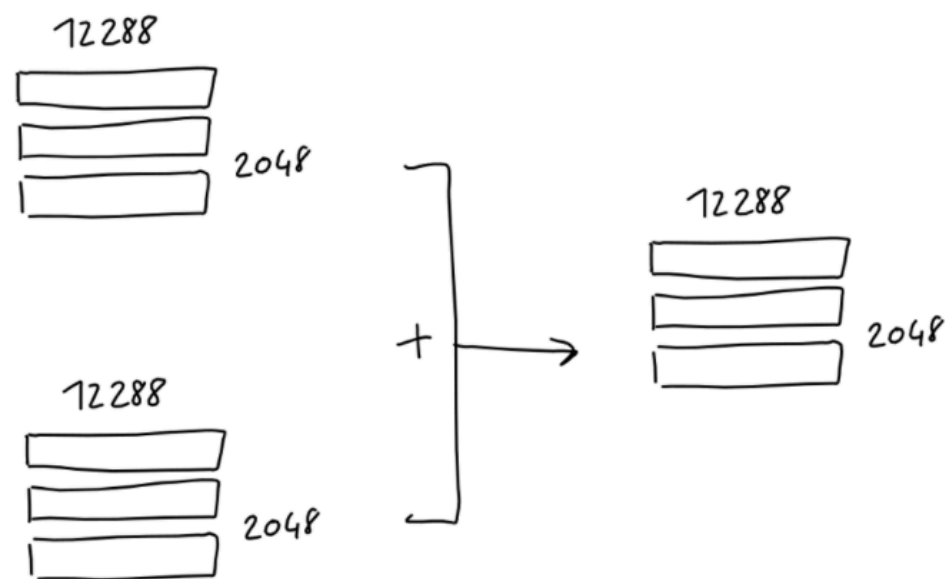
To encode the position of the current token in the sequence, the authors take the token's position (a scalar i , in $[0-2047]$) and pass it through 12288 sinusoidal functions, each with a different frequency.



The result is, for each token, a 12288 vector of numbers. Just as with the embeddings, we combine those vectors into a single matrix with 2048 rows, where each row is the 12288-column positional-encoding of a token in the sequence.

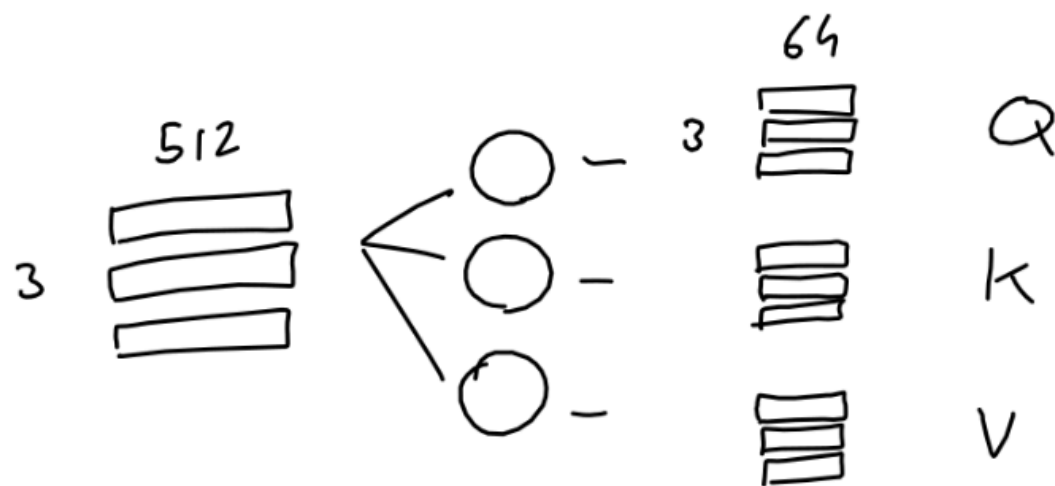


Finally, this sequence-positional-encodings matrix, having the same shape as the sequence-embeddings matrix, can simply be added to it.



Attention simplified

The model learns 3 linear projections, all of which are applied to the sequence embeddings. In other words, 3 weight matrices are learned which transform our sequence embeddings into three separate 3×64 matrices, each purposed for a different task.



Note: This (QK^T) is the only operation in GPT which operates *across* words in the sequence. It is the only operation where matrix rows interact.

$$\begin{array}{c} Q \\ 3 \end{array} \begin{bmatrix} \text{red line} \\ \text{green line} \\ \text{blue line} \end{bmatrix}_{512} \times \begin{array}{c} K^T \\ 512 \\ 3 \end{array} \begin{bmatrix} \text{red line} \\ \text{green line} \\ \text{blue line} \end{bmatrix} = \begin{array}{c} 0 \quad 1 \quad 2 \\ 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} 4.1 & -1 & 0.1 \\ . & . & 2.2 \\ . & 1.2 & -4. \end{bmatrix}$$

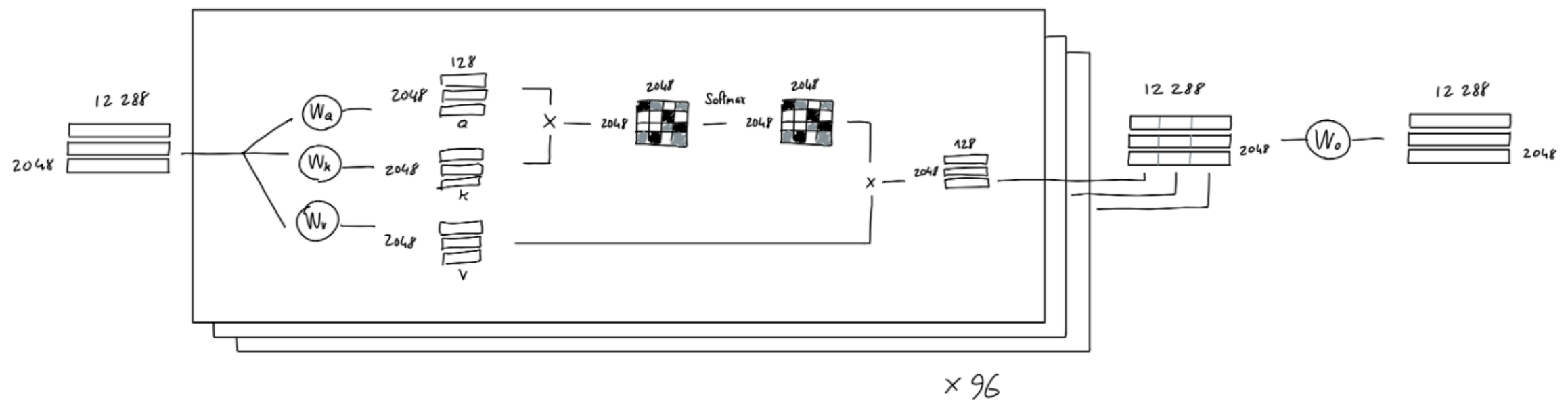
$$\text{Softmax}(QK^T) = \begin{array}{c} 0 \quad 1 \quad 2 \\ 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} 0.9 & 0.1 & 0.2 \\ . & . & 0.8 \\ . & 0.9 & 0.3 \end{bmatrix} \quad \begin{array}{c} 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \text{Diagram showing connections between nodes 0, 1, and 2} \end{array}$$

The third matrix ("values") is multiplied with this importance matrix, resulting in, for each token, a mix of all other token values weighted by the importance of their respective tokens.

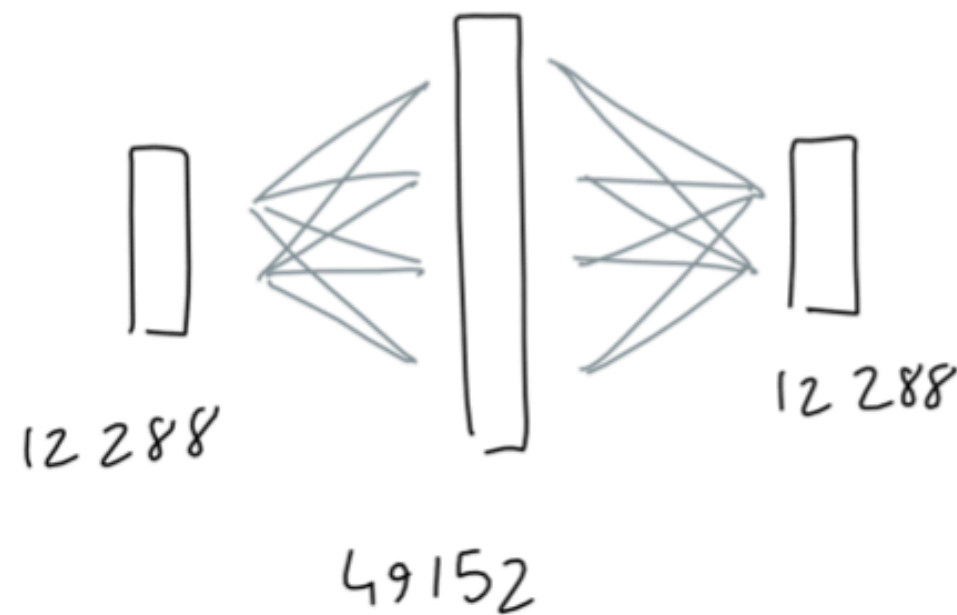
$$\begin{array}{c} \text{Softmax}(QK^T) \\ \begin{bmatrix} . & . & . \\ . & . & . \\ . & . & . \end{bmatrix} \end{array} \begin{array}{c} V \\ \begin{bmatrix} \text{red line} \\ \text{green line} \\ \text{blue line} \end{bmatrix}_3 \end{array}_{512} = 3 \begin{bmatrix} \text{red line} \\ \text{blue line} \\ \text{green line} \end{bmatrix}_{512} \quad \begin{array}{c} v \\ \text{Diagram showing multiplication of a 3x1 vector by a 3x1 vector} \end{array}$$

Mutli-head attention

The result of each attention head (a single 2048×128 matrix) are concatenated together, yielding a 2048×12288 matrix, which is then multiplied with a linear projection (which doesn't change the matrix shape), for good measure.

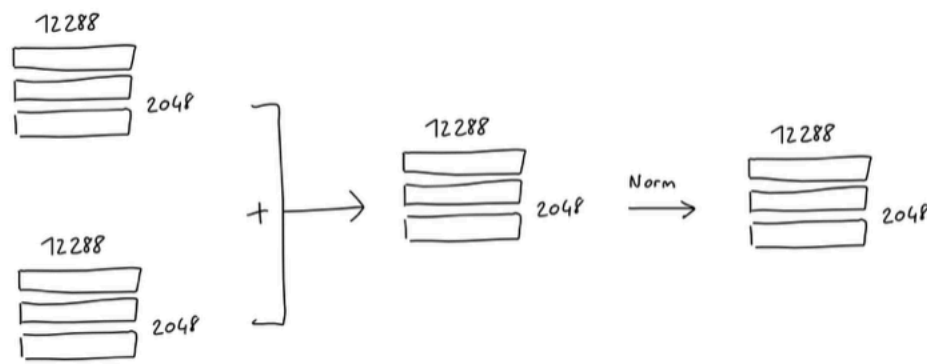


Feedforward



Add & Norm

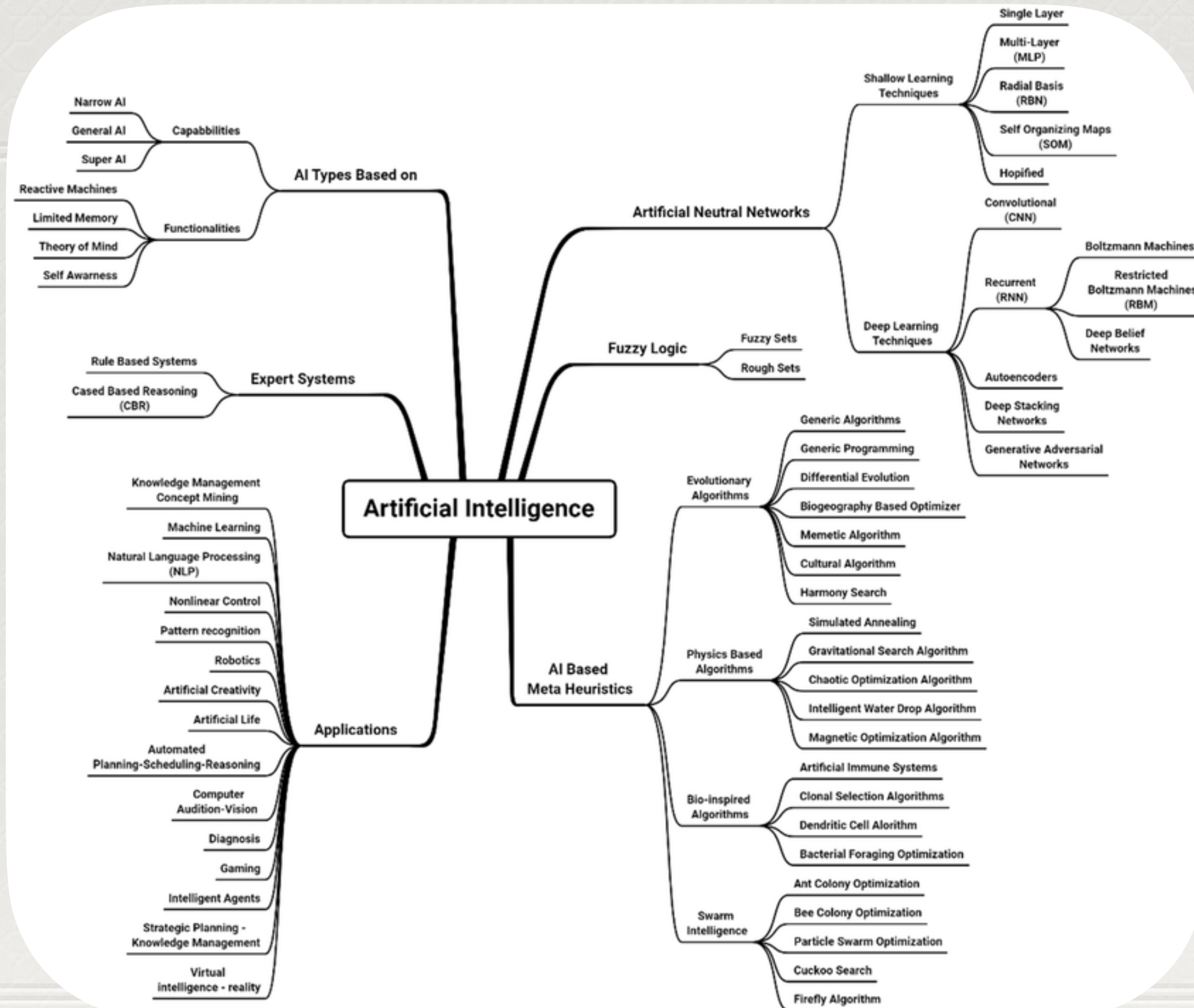
After both the Multi-Head attention and the Feed Forward blocks, the input of the block is added to its output, and the result is normalized. This is common in deep learning models (since ResNet).



Decoding



Other AI techniques



To conclude:

***AI = Data + data structure + algorithm +
probability + complexity***

*AI is everywhere, everyone appears to have become an “AI expert”.
Sometimes, AI experts may not understand the opaque box, but can succeed
in using AI in one context.*

*AI remains a probability. AI is as good as the data we
feed in. AI should be used responsibly, but with caution.*